

SHENGTING CAO

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RESEARCH INTERESTS

- AI-Driven Rehabilitation Devices Using AR/MR Technologies
- Neural Rendering and Generative AI
- Computer Vision and Deep Learning
- Data Science, LLM and RAG

EMPLOYMENT

Knox College, Galesburg, IL Tenure-Track Assistant Professor	Sep. 2025 – Present
University of Alabama, Tuscaloosa, AL Visiting Scholar	July 2025 – Sep. 2025
University of Alabama, Tuscaloosa, AL Research Assistant & Teaching Assistant	Aug. 2019 – May 2025

EDUCATION

The University of Alabama, Tuscaloosa, AL <i>Ph.D. in Electrical and Computer Engineering (ECE)</i> Dissertation: <i>Intelligent Treadmill Control and Holographic Rendering for Accessible Rehabilitation</i> <i>M.S. in Electrical and Computer Engineering (ECE)</i>	Aug. 2019 – May 2025 Dec. 2024
The University of Alabama, Tuscaloosa, AL <i>B.S. in Computer Science (CS)</i> <i>Minor: Advertising</i>	Jan. 2016 – May 2019

GRANTS AND PROPOSALS

Funded — External:

- **Sole Principal Investigator**, National Science Foundation Award #2553446. “ERI: Accessible XR System for Human Motion Monitoring, Reasoning, and Rendering with 3D Vision-Language Models.” Directorate for Engineering, Division of Electrical, Communications and Cyber Systems. \$199,736 to Knox College. Investigators: **Shengting Cao** (Single PI). Awarded July 10, 2026; period of performance July 15, 2026 – June 30, 2028.
- **Co-Principal Investigator**, National Science Foundation Award #2612358. “Collaborative Research: CyberTraining: Implementation: Medium: ACAI-Train: Scalable Instructor Training for Infusing AI and Advanced CI Concepts into Early Core Computing Courses.” Directorate for Computer and Information Science and Engineering, Office of Advanced Cyberinfrastructure. \$149,992 to Knox College. Investigators: David P. Bunde, Knox College (PI); Jaime Spacco, **Shengting Cao**, and Jurdana Masuma Iqrah (co-PIs). Awarded May 25, 2026; period of performance July 15, 2026 – June 30, 2030.

Under Review:

- **Co-Principal Investigator**, “GaitLane: A Smart-Glasses Human-Machine Interface for Straight and Symmetric Walking.” Samsung Global Research Outreach (GRO) Program 2026, Samsung Advanced Institute of Technology. Investigators: Mansoo Ko, University of Texas Medical Branch (PI); **Shengting Cao** (co-PI). Requested up to \$150,000, one year. Submitted August 2026; under review.

PUBLICATIONS

Patents:

- F. Hu, Y. Gan, **S. Cao**, and X. Wang, “Real-Time, Fine-Resolution Human Intra-Gait Pattern Recognition Based on Deep Learning Models,” U.S. Patent 12 592 097, Mar. 31, 2026.
- D. A. Brown, C.-Y. Li, M. Ko, **S. Cao**, X. Wang, F. Hu, Y. Gan, and L. Zhang, “Simulating a Split-Belt with a Single-Belt Treadmill,” U.S. Patent 12 370 404, Jul. 29, 2025.

Journals:

- J. Cui, **S. Cao**, J. Turner, B. Amer, S. Helus, J. Snow, J. Cronin, T. Sazonov, W. Lyu, J. Zhao, and F. Hu, “A Human–Machine System for Remote Rehabilitation: Cross-Platform Immersive Interaction Enabled by VR, AR, and Digital Twin Technologies,” *IEEE Transactions on Human-Machine Systems*, early access, pp. 1–11, 2026, doi: 10.1109/THMS.2026.3702529.
- X. Wu, **S. Cao**, and J. Gong, “Deep Learning-Based Spatio-Temporal Framework for Opioid Overdose Monitoring in Rural Alabama,” *Artificial Intelligence in Health*, vol. 3, no. 2, art. no. 025380076, 2026, doi: 10.36922/aih025380076.
- **S. Cao**, J. Zhao, F. Hu, and Y. Gan, “Real-Time, Free-Viewpoint Holographic Patient Rendering for Telerehabilitation via a Single Camera: A Data-Driven Approach with 3D Gaussian Splatting for Real-World Adaptation,” *IEEE Transactions on Visualization and Computer Graphics*, vol. 31, no. 10, pp. 7311–7323, 2025, doi: 10.1109/TVCG.2025.3544297.
- J. Gong, **S. Cao**, S. Korivand, and N. Jalili, “Reconstructing Human Gaze Behavior from EEG Using Inverse Reinforcement Learning,” *Smart Health*, vol. 32, art. no. 100480, 2024, doi: 10.1016/j.smhl.2024.100480.
- **S. Cao**, M. Ko, C.-Y. Li, D. Brown, X. Wang, F. Hu, and Y. Gan, “Single-Belt versus Split-Belt: Intelligent Treadmill Control via Microphase Gait Capture for Poststroke Rehabilitation,” *IEEE Transactions on Human-Machine Systems*, vol. 53, no. 6, pp. 1006–1016, 2023, doi: 10.1109/THMS.2023.3327661.
- X. Li, **S. Cao**, H. Liu, X. Yao, B. C. Brott, S. H. Litovsky, X. Song, Y. Ling, and Y. Gan, “Multi-Scale Reconstruction of Undersampled Spectral-Spatial OCT Data for Coronary Imaging Using Deep Learning,” *IEEE Transactions on Biomedical Engineering*, vol. 69, no. 12, pp. 3667–3677, 2022, doi: 10.1109/TBME.2022.3175670.
- H. Liu, **S. Cao**, Y. Ling, and Y. Gan, “Inpainting for Saturation Artifacts in Optical Coherence Tomography Using Dictionary-Based Sparse Representation,” *IEEE Photonics Journal*, vol. 13, no. 2, pp. 1–10, 2021, doi: 10.1109/JPHOT.2021.3056574.
- X. Chen, A. Miller, **S. Cao**, Y. Gan, J. Zhang, Q. He, R.-Q. Wang, X. Yong, P. Qin, B. H. Lapizco-Encinas, and K. Du, “Rapid *Escherichia coli* Trapping and Retrieval from Bodily Fluids via a Three-Dimensional Bead-Stacked Nanodevice,” *ACS Applied Materials & Interfaces*, vol. 12, no. 7, pp. 7888–7896, 2020, doi: 10.1021/acsami.9b19311.

Conference Publications:

- S. Kato* and **S. Cao**, “Tokenizing 2D Motion for Language Models: Geometric Supervision for Fine-Grained Rehabilitation Motion Understanding,” *IEEE/CVF Winter Conference on Applications of Computer Vision (WACV)*, 2027. Under review.
- **S. Cao**, X. Wu, Z. Zhao, M. Ye, and J. Gong, “Can Satellite Imagery See the Overdose Crisis? A Faithfulness-First Audit of GRPO-Trained Vision–Language Models on Drug and Social-Vulnerability Inference,” *AAAI Conference on Artificial Intelligence (AAAI), AI for Social Impact Track*, 2027. **Under review.**
- X. Wu, **S. Cao**, S. Li, and J. Gong, “SatBLIP: Context Understanding and Feature Identification from Satellite Imagery with Vision-Language Learning,” *arXiv:2604.14373*, 2026.
- Z. Zhao, **S. Cao**, and M. Ye, “SRVAU-R1: Enhancing Video Anomaly Understanding via Reflection-Aware Learning,” *arXiv:2602.01004*, 2026.
- X. Zhu, X. Guo, **S. Cao**, S. Li, and J. Gong, “StructuGraphRAG: Structured Document-Informed Knowledge Graph for Retrieval-Augmented Generation,” in *Proc. AAAI Symposium Series*, vol. 4, no. 1, 2024, pp. 242–251.

- X. Wu, **S. Cao**, H. Y. Lee, and J. Gong, “Let Every Voice Be Heard: Developing a Cost-Effective Community Sampling Frame in Rural Alabama to Combat COVID-19,” in *Proc. IEEE/ACM Conf. on Connected Health: Applications, Systems and Engineering Technologies (CHASE)*, 2022, pp. 174–175. (poster)
- **S. Cao**, X. Yao, N. Koirala, B. Brott, S. Litovsky, Y. Ling, and Y. Gan, “Super-Resolution Technology to Simultaneously Improve Optical & Digital Resolution of Optical Coherence Tomography via Deep Learning,” in *Proc. 42nd Annu. Int. Conf. IEEE Engineering in Medicine & Biology Society (EMBC)*, 2020, pp. 1879–1882, doi: 10.1109/EMBC44109.2020.9175777.
- M. V. Fedewa, K. Sullivan, C. J. Holmes, B. Hornikel, **S. Cao**, Y. Gan, and M. R. Esco, “Test-Retest Reliability of Total Body Volume Derived from a Single 2-Dimensional Digital Image,” *Medicine & Science in Sports & Exercise*, vol. 52, no. 7S, p. 869, 2020. (abstract)
- K. Sullivan, C. J. Holmes, B. Hornikel, **S. Cao**, Y. Gan, M. R. Esco, and M. V. Fedewa, “Validity of a 3-Compartment Body Composition Model Derived from a Single 2-Dimensional Digital Image,” *Medicine & Science in Sports & Exercise*, vol. 52, no. 7S, p. 870, 2020. (abstract)

Edited Book:

- F. Hu, **S. Cao**, and X.-S. Hei, Eds., *Metaverse Platform Implementation: AI, Security, and Applications*. Boca Raton, FL, USA: CRC Press, 2025.

Book Chapters:

- J. Cui and **S. Cao**, “Virtual Reality Medical Technology in the Metaverse: A Digital Twin Approach and Cybersecurity Considerations,” in *Metaverse Platform Implementation: AI, Security, and Applications*, F. Hu, **S. Cao**, and X.-S. Hei, Eds. Boca Raton, FL, USA: CRC Press, 2025, pp. 249–264.
- **S. Cao** and X. Wu, “Neural Rendering Technology in VR and AR,” in *Metaverse Platform Implementation: AI, Security, and Applications*, F. Hu, **S. Cao**, and X.-S. Hei, Eds. Boca Raton, FL, USA: CRC Press, 2025, pp. 3–27.
- A. Cantu, W. Lyu, and **S. Cao**, “High-Bandwidth Communication Technologies for the Metaverse,” in *Metaverse Platform Implementation: AI, Security, and Applications*, F. Hu, **S. Cao**, and X.-S. Hei, Eds. Boca Raton, FL, USA: CRC Press, 2025, pp. 81–95.

* *Knox College undergraduate co-author.*

TEACHING EXPERIENCE

Courses Taught, Knox College (Assistant Professor):

- | | |
|--|-------------|
| CS 343 Neural Artificial Intelligence | Spring 2026 |
| <ul style="list-style-type: none"> • Designed a project-based course in which student teams build and commercialize real AI products, teaching AI-assisted development with agentic coding tools (e.g., Claude Code) as a first-class engineering skill • Taught state-of-the-art large language models and diffusion models | |
| CS 127 Computing, Technology, and Society | Spring 2026 |
| <ul style="list-style-type: none"> • Taught Python programming and computing fundamentals to a no-prerequisite, all-majors audience | |
| CS 317 Artificial Intelligence | Winter 2026 |
| <ul style="list-style-type: none"> • Designed and taught search algorithms, knowledge representation, probabilistic reasoning, machine learning, RL and LLM | |
| CS 142 Program Design and Methodology (Java II) | Winter 2026 |
| <ul style="list-style-type: none"> • Taught object-oriented programming in Java and foundational data structures such as stacks | |
| CS 208 Programming Languages | Fall 2025 |
| <ul style="list-style-type: none"> • Designed and taught functional programming with Haskell, including compilers, parsers, and related concepts, enabling students to design and implement their own programming languages | |

Teaching Prior to Knox, University of Alabama:

Teaching Assistant in ECE

ECE492/494 Capstone Design I/II (twice/week)

May 2020 – May 2025

- Guided senior design groups in defining research problems and developing prototype solutions, managing an annual timeline to achieve key milestones

ECE 409/ECE509: Communication Labs (once/week)

Spring 2024, Spring 2021

- Instructed students to familiarize themselves with *MATLAB* built-in functions and model *multi-path propagation*, *modulation*, and *signal constellations* for communication systems

ECE 380: Digital Logic (once/week, three sections/semester)

Fall 2023, Spring 2023, Fall 2022

- Instructed students in fundamental concepts of digital design, including standard IC design and *FPGA* development using *VHDL*

Teaching Assistant in CS

Data Science Summer Bootcamp (every weekday for one month)

Summer 2022

- Assisted students in becoming proficient with *RapidMiner* for processing CSV files and applying machine learning algorithms such as *SVM* and *Random Forest*

Undergraduate Tutor in CS

CS100/CS101: Computer Science I/II for Major (twice/week)

May 2018 – May 2019

- Assisted students in becoming proficient with C and C++ compilers, covering programming concepts such as *loops*, *recursion*, *sorting*, *algorithmic complexity*, *data structures*, and *object-oriented programming*

CS201: Data Structures and Algorithms (twice/week)

May 2018 – May 2019

- Helped students understand and implement algorithmic concepts including *Divide-and-Conquer*, *sorting algorithms*, *hash tables*, *dynamic programming*, *greedy algorithms*, and *graph algorithms*

Student Research Mentorship:

- **Honors Thesis Advisor and Committee Chair (2025-2026):** Richie Dix, “Integrating and Fine-tuning Single-Image-to-3D Model for 4D XR Human Visualization,” B.A. with College Honors, Knox College, defended June 2, 2026. Funded by a Richter Grant, Vovis Center. Committee: S. Cao (chair), D. P. Bunde, O. Forsberg, B. Xiang (Vanderbilt University).
- **Supervised Undergraduate Research, Summer 2026 (four students):** Sabina Stalbekova (Richter Grant), Mian Hissan Tahir (Knox Summer Scholar), Ahsan Tariq (Richter Grant), and Seishin Kato (Richter Grant). Kato’s project led to a first-authored paper submission to WACV 2027 (under review).
- **Student Mentorship, University of Alabama (2020–2025):** mentored 13 undergraduate and graduate students on senior design capstone projects and on master’s thesis research — Nguyen Hung Nguyen, Colin Aderon, Sarah Scott, Wesley Junkins, Adam Tompkins, Trent Boughner, Abdullah Alsubaei, Andres Aguilar Calderon, Joe Dalpoas, Tyler Cruise, Jacob Tomberg, Jessica Tashbin, and Nolan McKivergan.

Fields Prepared to Teach:

- **Computer Science:** introductory programming; data structures and algorithms; programming languages; artificial intelligence; machine learning; deep learning; computer vision; data science.
- **Electrical and Computer Engineering:** digital logic; microcomputers; communication systems; computer architecture.

HONORS AND AWARDS

- **NSF Innovation Corps Program Certification.** Issued by the National Science Foundation for entrepreneurial leadership training; interviewed over 120 physical and occupational therapists to validate a startup business model (2021)
- **UTMB Technology Commercialization Program Seed Fund.** Awarded travel and living expenses for Splicer software development at the University of Texas Medical Branch, Galveston, Texas (2021)
- **Conference & Research Support Funding.** Awarded \$300 for the invited talk at the University of West Alabama Symposium (Fall 2020)

- **First place in the Google Earth Engine Challenge at the Machine Learning for Science Hackathon Competition.** Awarded by the Alabama Cyber Initiative of the University of Alabama and Data Science Initiative of Brown University (2019)
- **Collaborative Event Grant (SUPE Store Sponsored).** Awarded \$5,000 for the Spring Festival event (2019)

SERVICE

Institutional Service, Knox College:

- **Faculty Adviser, Vietnamese Student Club, Knox College (Spring 2026 – Present)**

Professional Service:

- **Book Co-editor: Metaverse Platform Implementation: AI, Security, and Applications, CRC Press, 2025**
- **Reviewed papers:**
 - Medical Image Analysis, 2026
 - IEEE Transactions on Systems, Man and Cybernetics: Systems, 2026
 - Scientific Reports (Springer Nature), 2026
 - Artificial Intelligence in Health, 2026
 - Health Informatics Journal (SAGE), 2026
 - AI for Society (MDPI), 2026
 - International Journal of Telemedicine and Applications (IJTA), 2021 – 2026
 - IEEE International Symposium on Mixed and Augmented Reality (ISMAR), 2025

Community Service:

- **Alabama K-12 Robotics Competition Volunteer (2024):** Assisted with power and competition environment setup
- **AAPI Heritage Month Event Volunteer (2024):** Introduced and showcased Chinese board games; managed food distribution
- **Chinese School Teacher (CST):** Instructed Chinese language classes for elementary school students
- **ACSSUA (Association of Chinese Students and Scholars at the University of Alabama):** Served in different roles such as President, Vice President, Social Media Ambassador, and Spring Festival Event Volunteer for 7 years. Organized cultural events, managed COVID-19 relief efforts, and handled social media outreach
- **MATHCOUNTS West Alabama Regional Competition Volunteer:** Demonstrated 3D optical scanning of fingerprints and Virtual Reality prototypes to high school students
- **AI's Pals Volunteer Mentor:** Tutored 1st-grade students in English and Math weekly at Matthews Elementary School